

INTERNET ROUTER

Complete Setup, Configuration & Troubleshooting Guide

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Applicable Models	Broadband / DSL / Fiber / Cable Routers
Audience	Home Users & Small Business Owners
Language	English

This manual provides step-by-step instructions for setting up, configuring, securing, and troubleshooting your internet router. Whether you are a first-time user or an experienced network administrator, this guide will help you get the best performance from your device.

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1. Safety Information

Before installing or using your router, please read the following safety precautions carefully:

- Place the router in a well-ventilated area. Do not block ventilation openings.
- Keep the device away from liquids, extreme heat, and direct sunlight.
- Do not attempt to open or repair the router yourself — this voids the warranty.
- Use only the power adapter supplied with the device.
- Unplug the router during lightning storms or when not in use for extended periods.
- Keep out of reach of small children to avoid accidental damage.

■ **WARNING:** Improper use may cause fire, electric shock, or product damage. Always follow local electrical regulations.

2. Package Contents & Hardware Overview

2.1 Package Contents

Item	Quantity
Wireless Router Unit	1
Power Adapter (12V DC)	1
Ethernet Cable (Cat5e, 1m)	1
Quick Start Guide	1
Mounting Screws (optional)	2

2.2 LED Indicators

LED	Color	Status	Meaning
Power	Blue	Solid	Router is on and operating normally
Internet	Blue/Red	Solid Blue	Internet connection established
	Blue/Red	Solid Red	No internet connection detected
Internet	Blue	Blinking	Wireless activity on 2.4 GHz band
Wi-Fi 5G	Blue	Blinking	Wireless activity on 5 GHz band

LAN 1–4	Blue	Solid	Wired device connected
WPS	Blue	Blinking	WPS pairing in progress

2.3 Rear Panel Ports

Port/Button	Description
WAN Port	Connect the incoming internet cable (from modem or ISP) here
LAN 1–4	Connect wired devices such as PCs, smart TVs, or gaming consoles
USB Port	Connect USB storage for network file sharing or 3G/4G backup
Power Jack	Connect the included 12V power adapter
Reset Button	Hold 10 seconds to restore factory defaults
WPS Button	Press to enable Wi-Fi Protected Setup pairing
Power Switch	Toggle router on/off

3. Installation & Physical Setup

3.1 Choosing a Location

For the best wireless coverage and performance, choose an installation location that meets these criteria:

- Central location in your home or office for even signal distribution.
- Elevated position (e.g., a shelf) rather than on the floor.
- Away from walls, metal objects, microwaves, and cordless phones that cause interference.
- Near a power outlet and the modem or ISP connection point.
- Avoid enclosed spaces such as cabinets or closets.

3.2 Connecting the Hardware

Follow these steps in order:

Step 1	Power off your modem (the device provided by your ISP).
Step 2	Connect one end of the Ethernet cable to your modem's LAN port and the other end to the router's WAN (Internet) port.
Step 3	Plug the power adapter into the router's DC jack, then connect it to a wall outlet.
Step 4	Switch the router ON using the Power button.
Step 5	Power on your modem and wait 60 seconds for it to reconnect to your ISP.
Step 6	Wait for the Internet LED on the router to turn solid blue, indicating a successful connection.

✓ NOTE: If the Internet LED remains red after 2 minutes, verify the cable is securely plugged into the WAN port and that your modem is active.

4. Accessing the Router Admin Panel

The router's web-based administration panel allows you to configure all settings. To access it:

1. Connect your computer to the router via Ethernet cable or Wi-Fi.
2. Open a web browser (Chrome, Firefox, Edge, Safari).
3. In the address bar, type the router's default gateway IP address:

`http://192.168.1.1` or `http://192.168.0.1`

4. You will see the login page. Enter the default credentials:

Field	Default Value
Username	admin
Password	admin (or blank — check label on router bottom)

■ **IMPORTANT:** Change the default admin password immediately after first login to prevent unauthorised access.

5. Initial Configuration

5.1 Internet Connection Setup

Navigate to **Basic > Internet** in the admin panel. Select the connection type provided by your ISP:

Connection Type	When to Use
DHCP (Dynamic IP)	Most broadband / cable connections (most common)
PPPoE	DSL connections requiring a username and password from ISP
Static IP	When your ISP has assigned you a fixed IP address
L2TP / PPTP	Used by some corporate or international ISP configurations

After selecting the correct type and entering any required credentials, click **Save** and wait for the router to reconnect.

5.2 Wi-Fi Network Setup

Navigate to **Wireless > Settings**. Configure the following for each band (2.4 GHz and 5 GHz):

Setting	Recommended Value	Notes
Network Name (SSID)	Your chosen name	Avoid using personal info
Security Mode	WPA3-Personal or WPA2	WPA3 preferred if supported
Password	12+ characters	Mix letters, numbers, symbols
Channel Width (2.4G)	20 MHz or Auto	Reduces interference
Channel Width (5G)	80 MHz or Auto	Best for HD streaming
Band Steering	Enabled	Auto-connects to best band

5.3 Changing the Admin Password

Go to **Administration > Management > Admin Password**. Enter your current password, then set a strong new password (minimum 10 characters). Click **Save**. Store this password in a secure location.

6. Advanced Settings

6.1 DHCP Server Settings

The DHCP server automatically assigns IP addresses to devices on your network. Navigate to **Advanced > LAN > DHCP**:

- Default range: 192.168.1.100 – 192.168.1.200 (suitable for most homes).
- DHCP Lease Time: 1440 minutes (24 hours) is recommended.
- To assign a fixed IP to a specific device, use DHCP Reservation by binding its MAC address.

6.2 Port Forwarding

Port forwarding allows external internet traffic to reach a specific device on your local network. This is commonly used for gaming servers, remote desktop, or IP cameras. Navigate to **Advanced > NAT > Port Forwarding**:

- Service Name: A descriptive label (e.g., 'Minecraft Server').
- Protocol: TCP, UDP, or Both — check your application's documentation.
- External Port: The port number incoming traffic will use.
- Internal IP: The local IP address of the target device.
- Internal Port: Usually the same as external port.

✓ TIP: Assign a static DHCP reservation to the target device before configuring port forwarding so its IP does not change.

6.3 Quality of Service (QoS)

QoS prioritises certain types of network traffic to ensure smooth performance for time-sensitive applications such as video calls and online gaming. Go to **Advanced > QoS**:

Priority Level	Recommended Traffic Types
Highest	VoIP calls, video conferencing (Zoom, Teams)
High	Online gaming, live streaming
Medium	Web browsing, email, social media
Low	File downloads, cloud backups, updates

6.4 Guest Network

A guest network lets visitors connect to the internet without accessing your main local network, protecting your files and devices. Navigate to **Wireless > Guest Network**:

- Enable Guest Network and set a separate SSID (e.g., 'HomeGuest').
- Set a strong password for the guest network.
- Enable 'Access Point Isolation' to prevent guests from communicating with each other.
- Optionally set a bandwidth limit for guest users.

6.5 Parental Controls

Navigate to **Advanced > Parental Controls** to manage internet access for specific devices:

- Add devices by MAC address to create individual profiles.
- Set allowed browsing hours with scheduled access windows.
- Block specific websites or categories (adult content, gaming, social media).
- Use Safe Search enforcement for Google and Bing.

7. Security Best Practices

Securing your router is critical to protecting all devices on your network. Follow these best practices:

- **Change Default Admin Credentials:** Never leave the router on factory default username/password.
- **Use WPA3 or WPA2 Encryption:** Older protocols (WEP, WPA) are easily cracked.
- **Disable WPS:** WPS (Wi-Fi Protected Setup) has known security vulnerabilities. Disable it under Wireless settings.
- **Disable Remote Management:** Unless needed, turn off access to the admin panel from the internet.
- **Enable Firewall:** Keep SPI Firewall enabled under Advanced > Security.
- **Disable UPnP:** Universal Plug and Play can expose your network. Disable if not needed.
- **Use a Strong Wi-Fi Password:** At least 12 characters combining upper/lowercase, numbers, and symbols.
- **Regularly Update Firmware:** Updates patch security vulnerabilities (see Section 8).
- **Monitor Connected Devices:** Check the device list regularly under Status > Attached Devices.

■ **WARNING:** An unsecured router can allow neighbours or malicious actors to use your internet connection, intercept your data, or attack your devices.

8. Firmware Updates

Firmware is the software that runs on your router. Keeping it updated ensures you have the latest security patches and performance improvements.

Automatic Update (Recommended)

- Log in to the admin panel.
- Go to Administration > Firmware Update.
- Enable 'Auto Update' and set a preferred update schedule (e.g., 3:00 AM on Sundays).

Manual Update

- Visit the router manufacturer's official support website.
- Search for your router model and download the latest firmware file (.bin or .trx).
- In the admin panel, go to Administration > Firmware Update > Manual Update.
- Browse to the downloaded file and click Upload.
- Do NOT power off the router during the update. Wait for the process to complete (approx. 3–5 minutes).
- The router will reboot automatically when finished.

✓ NOTE: After a firmware update, your settings are usually preserved. However, it is good practice to export your configuration backup beforehand via Administration > Backup Settings.

9. Troubleshooting

Problem	Possible Cause	Solution
No internet access (Internet LED red)	Modem not connected or ISP outage	Check WAN cable. Reboot modem. Verify ISP service status.
Cannot connect to Wi-Fi	Wrong password or signal too weak	Confirm password. Move closer to router. Check SSID is visible.
Slow internet speed	Congestion, interference, or outdated firmware	Run speed test at fast.com. Change Wi-Fi channel. Update firmware.
Admin panel not loading	Wrong IP or browser cache	Try 192.168.0.1. Clear browser cache. Use a different browser.
Devices keep disconnecting	IP conflict or DHCP issue	Enable DHCP reservation. Check for IP address conflicts.
Forgot admin password	Password changed and forgotten	Reset router to factory defaults (hold Reset button 10 seconds).
Wi-Fi range is poor	Physical obstructions or wrong channel	Reposition router. Use a Wi-Fi extender for larger spaces.
One device cannot connect	MAC filter blocking it	Check MAC Address Filter list in Wireless settings.

Factory Reset Procedure

If all else fails, a factory reset returns the router to its original out-of-box state. **This erases all custom settings.**

- Ensure the router is powered on.
- Locate the Reset button (usually a small pinhole on the rear panel).
- Using a pin or paperclip, hold the Reset button for 10 seconds.
- Release when the Power LED blinks rapidly.
- Wait 60 seconds for the router to fully restart.
- Reconfigure the router from scratch using this manual.

10. Technical Specifications

Specification	Details
Wireless Standards	IEEE 802.11a/b/g/n/ac/ax (Wi-Fi 6)
Frequency Bands	2.4 GHz and 5 GHz (Dual-Band)
Maximum Wireless Speed	AX3000: Up to 574 Mbps (2.4G) + 2402 Mbps (5G)
LAN Ports	4 x Gigabit Ethernet (10/100/1000 Mbps)
WAN Port	1 x Gigabit Ethernet
USB Port	1 x USB 3.0
Antennas	4 x External High-Gain Antennas
Security	WPA3, WPA2, WPA, WEP, SPI Firewall, VPN
Power Supply	12V / 2A DC
Operating Temperature	0°C – 40°C (32°F – 104°F)
Storage Temperature	-40°C – 70°C
Humidity	10% – 90% non-condensing
Dimensions (L x W x H)	220 x 140 x 35 mm
Weight	Approx. 420 g (without adapter)
Certifications	FCC, CE, RoHS

11. Warranty & Support

This router is covered by a **2-year limited hardware warranty** from the date of purchase. The warranty covers manufacturing defects under normal use conditions.

Warranty does NOT cover:

- Physical damage caused by misuse, dropping, or liquid exposure.
- Damage from use of non-approved power adapters.
- Products with tampered or removed serial numbers.
- Issues arising from unauthorised firmware or hardware modifications.

Contacting Support

Support Channel	Details
Website	www.routersupport.example.com
Email	support@routersupport.example.com
Phone (Toll-Free)	+1-800-555-0199 (Mon–Fri, 9AM–6PM EST)
Live Chat	Available on website, Mon–Sun 8AM–10PM EST
Community Forum	community.routersupport.example.com

When contacting support, please have the following ready: router model number, serial number (on the label under the device), firmware version, and a description of the issue.